

LAB - In class

Topic 3: Complex SQL Queries



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In-class assignment

Task 1: Use University database (run UniversityDB_script file on moodle to create the database). Utilize the SQL editor of SQL Management Studio or Azure Data Studio to write the following queries:

1. Using CTE, for each department, find all the top 2 students based on the average grade. Include all the students that tied for the top 2.
2. Find all courses that directly or indirectly depend on course CS07 as a prerequisite.
3. Create the following tables with necessary constraints (PK, FK, the constraints on value domain):
 - a. TeachingCapacity
 - i. instructor_id: varchar(9) not null primary key, reference to the Instructor table.
 - ii. course_id: varchar(9) not null primary key, reference to the Course table.
 - iii. nb_year: int (>0).
4. Insert data into the above table using a query.
5. Update the grade_ABC column of the GradeReport table based on the values in the grade_100 column:
 - a. 90-100: A,
 - b. 80-89: B,
 - c. 70-79: C,
 - d. 65-69: D.
 - e. 50-65: E,
 - f. <50: F
6. Find the ID and name of each student, along with the average grade (grade_100) of the sections they have passed, where a passed section is defined as having a grade_100 of 50 or higher.
7. Retrieve the code and name of ALL students, along with the count of courses they have passed, where a passed course is defined as having a grade_100 of 50 or higher.
8. For each department, find the student(s) with the highest average grade across all enrolled sections. Return the department_id, department name, student_id, student name and average grade.
9. List students whose average grade is higher than the average of all students in their department.
10. Create the StudentStatistics table with the primary key and all necessary foreign keys to summarize student information. The table should include the following columns: student_id, student_name, school_year, semester, and nb_credits (representing the total number of credits for the courses enrolled in each semester of each school year). After creating the table, populate it using data extracted from the existing database.